

B. Tech. PROJECT REPORT (Mid-Semester Evaluation)

WIDEBAND MIMO ANTENNA WITH METASURFACE

Phase I: CPW-fed UWB Antenna with a Split-Ring-Resonator Metasurface Reflector



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Abstract

Ultra-wideband (UWB) communication and multiple-input multiple-output (MIMO) technology together support high-speed, low-latency wireless links. However, conventional printed planar monopoles radiate bidirectionally, yielding low forward gain ($\sim 2\text{--}4$ dBi). This project develops a high-gain, low-profile wideband MIMO antenna integrated with a metasurface reflector. This mid-semester report presents Phase I: design, optimization, and characterization of a single coplanar waveguide (CPW) fed decagonal monopole antenna on a $50 \times 50 \times 1.6$ mm FR-4 substrate backed by a copper-backed split-ring-resonator (SRR) metasurface. Through a two-stage parametric optimization in CST Studio Suite 2019—varying the ground edge L_g from $y = -20$ mm to $y = -7$ mm and patch circumradius R from 4 mm to 15 mm—the antenna achieves $|S_{11}| \leq -10$ dB from 2.16 GHz to 15.73 GHz (fractional bandwidth of 151.7 %), comprehensively covering the 2.1 GHz to 15 GHz spectrum with an IEEE gain of 2.9–4.9 dBi. A 6×5 double-SRR metasurface placed 3.9 mm behind the antenna provides in-phase reflection to substantially boost forward broadside gain without impedance detuning. Phase II will expand this design into an orthogonal MIMO array.

Introduction

Background and Motivation

High-speed wireless links for radar, imaging, and sensing demand multi-gigahertz bandwidths covering 2.1 GHz to 15 GHz. MIMO antenna architectures multiply channel capacity without additional transmit power by exploiting multipath scattering [1], [2]. Planar printed monopoles are favored for wideband systems due to their broad impedance bandwidth [3], [4]. A CPW feed keeps the signal strip and ground planes on the same side, eliminating vias and leaving the rear bare [5]. However, planar monopoles radiate bidirectionally, leading to low forward gain ($\sim 2\text{--}4$ dBi) [6]. While a metal plate requires $\lambda_0/4$ separation to prevent destructive cancellation, an artificial magnetic conductor (AMC) metasurface achieves in-phase reflection, enabling an ultra-thin profile (3.9 mm gap) [7], [8].

Problem Statement and Objectives

The objective is to design a compact CPW-fed planar decagonal monopole operating across 2.1 GHz to 15 GHz, enhance forward broadside gain using an SRR metasurface reflector at an ultra-thin 3.9 mm air gap, and extend the configuration into a high-isolation multi-port MIMO system [9].

- Design and optimize a CPW-fed decagonal monopole antenna achieving $|S_{11}| \leq -10$ dB from 2.1 GHz to 15 GHz.
- Model and characterize an SRR unit cell to achieve AMC in-phase reflection characteristics near resonance.
- Integrate a 6×5 metasurface array at an ultra-thin 3.9 mm air gap to boost forward gain without detuning matching.
- Develop a 4-port orthogonal MIMO array ensuring high inter-port isolation and low envelope correlation.
- Fabricate prototypes on FR-4 substrate and validate simulated reflection, isolation, and radiation characteristics experimentally.

Literature Review

Recent literature on metasurface-assisted antennas addresses single-element gain enhancement and multi-port MIMO decoupling. In single-element designs, Al-Gburi et al. [10] and Hussain et al. [11] demonstrated 4.5–6 dB gain boosts using FSS reflectors, but required large air gaps (9–10 mm). In MIMO implementations, Hasan et al. [12] used a copper-backed SRR metasurface at a 12 mm air gap to achieve > 15.5 dB isolation and 8.3 dBi gain, but covered only 3.08–7.75 GHz. Wu et al. [13] integrated a polarization-conversion metasurface for 5 GHz WBAN MIMO. Table 1 benchmarks representative literature against this work.

Table 1. Comparison of recent wideband / MIMO antennas with metasurface reflectors against this work.

Reference	Antenna Type	Band (GHz)	Metasurface / Reflector	Air Gap	Gain Enhancement	Isolation
Al-Gburi (2022) [10]	1-port CPW ring	3.08–11.5	19×19 cross-loop FSS	10 mm ($0.10\lambda_d$)	$6.7 \rightarrow 11.5$ dBi	—
Hussain (2023) [11]	1-port UWB disc	3.4–10.6	8×8 slotted FSS	9 mm ($0.10\lambda_d$)	$2.2 \rightarrow 8.4$ dBi	—
Hasan (2022) [12]	4-port square patch	3.08–7.75	10×10 SRR AMC (backed)	12 mm ($0.12\lambda_d$)	$5.4 \rightarrow 8.3$ dBi	> 15.5 dB
Wu (2023) [13]	2-port CPW monopole	4.76–6.77	Polarization-conversion MS	Integrated	7.95 dBic	> 19.8 dB
This Work	1-port decagon ($\rightarrow$ MIMO)	2.16–15.73	6×5 double-SRR AMC	3.9 mm ($0.03\lambda_d$)	≈ 2.9 –4.9 dBi $\rightarrow$ pending	> 15 dB (target)

Research Gap

Existing literature reveals three principal gaps: (1) reflectors behind UWB monopoles use large gaps (9–20 mm, ~ 0.10 – $0.20\lambda_0$), with no reported design under a sub-4 mm profile; (2) prior works rarely benchmark metasurface gain against a metal plate at the same spacing; and (3) reported metasurface MIMO antennas cover narrower sub-bands rather than the full 3.1–10.6 GHz UWB standard. This project bridges this gap with an ultra-thin 3.9 mm profile and full UWB MIMO coverage.

Methodology / Proposed Work

1. Overall Workflow and Methodology

The sequential workflow is shown in Fig. 1: setting up the baseline CPW radiator, optimizing ground edge L_g for matching, sweeping patch radius R for bandwidth, synthesizing the optimized monopole, integrating an SRR metasurface at 3.9 mm air gap, and 4-port MIMO extension.

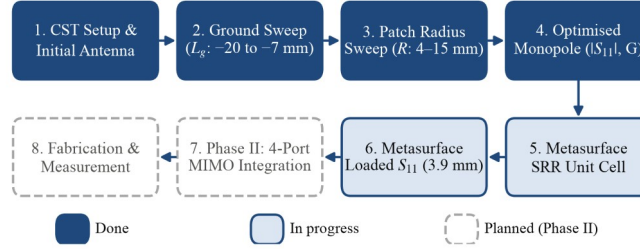


Fig. 1. Design and optimization workflow of the wideband antenna and metasurface.

2. CST Microwave Studio Simulation Setup

Simulations used CST Studio Suite 2019 Frequency-Domain Solver (0–18 GHz) with adaptive tetrahedral meshing. The antenna is modeled on 50×50 mm FR-4 ($\epsilon_r = 4.3$, $\tan \delta = 0.025$, $h_s = 1.6$ mm, 35 μ m copper). The 50 Ω CPW feed has a 3.0 mm signal strip and two coplanar grounds separated by 0.5 mm slots, excited by a waveguide port.

3. Initial Antenna Design and Baseline S_{11} Performance

The initial radiator has a decagonal patch ($R = 15$ mm, center $x_p = 8$ mm) with CPW grounds at $y = -20$ mm ($L_g = -20$ mm, Fig. 2(a)), leaving a large 13.73 mm gap to the patch. Weak capacitive coupling across this gap kept baseline $|S_{11}|$ above -10 dB (-3 to -8 dB) across almost the entire spectrum (Fig. 2(b)), failing wideband requirements.

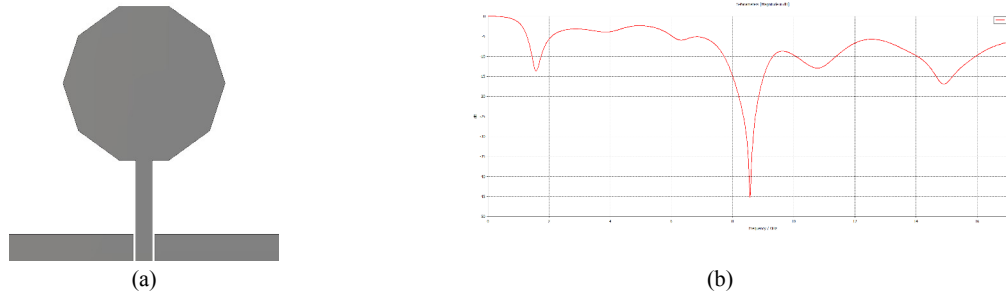


Fig. 2. Initial antenna model: (a) initial CPW decagonal geometry ($L_g = -20$ mm), and (b) simulated baseline reflection coefficient $|S_{11}|$ showing poor matching across 2.1–15 GHz.

4. Parametric Optimization of Ground Patch Length (L_g Sweep)

Ground length L_g was swept from $y = -20$ mm to $y = -7$ mm (10 steps, $R = 15$ mm). Narrowing gap p from 13.73 to 0.73 mm strengthens capacitive coupling, shifting the first resonance from 1.6 to 2.73 GHz and deepening it below -30 dB (Fig. 3(a)). Only $L_g = -7$ mm maintains $|S_{11}| \leq -10$ dB across the band, establishing $L_g = -7$ mm as optimal.

5. Parametric Optimization of Radiating Patch Radius (R Sweep)

With $L_g = -7$ mm, patch circumradius R was swept from 4 to 15 mm (13 values). For $R = 4$ –8 mm, modes resonate above 6 GHz. Increasing R enlarges electrical volume, shifting the lower cutoff downward as $f_L \approx 7.2 / (L + r + p)$ [3], [4]. At $R = 15$ mm (Fig. 3(b)), multiple resonant modes coalesce into a continuous wideband across 2.1–15 GHz.

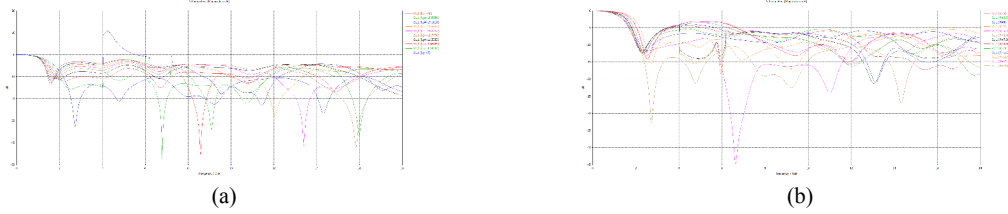


Fig. 3. Parametric sweeps: (a) simulated $|S_{11}|$ vs. ground edge L_g from $y = -20$ to -7 mm, and (b) simulated $|S_{11}|$ vs. patch radius R from 4 to 15 mm (optimal: $R = 15$ mm, $L_g = -7$ mm).

6. Optimized Monopole Antenna Performance

The optimized standalone antenna ($R = 15$ mm, $L_g = -7$ mm, $p = 0.73$ mm, Fig. 4(a)) maintains $|S_{11}| \leq -10$ dB from 2.16 to 15.73 GHz (FBW = 151.7 %), fully covering 2.1 GHz to 15 GHz (Fig. 4(b)). Four distinct resonances occur at 2.73 GHz, 4.78 GHz, 9.23 GHz, 14.32 GHz, with a peak return loss of 32.65 dB at 2.73 GHz. Simulated IEEE gain is 2.9–4.9 dBi across 3.1–10.6 GHz (peak 5.09 dBi near 13.5 GHz).

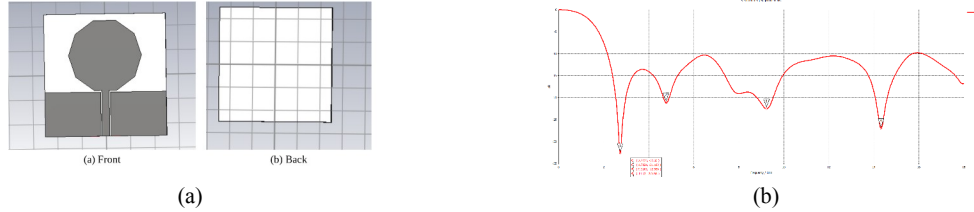


Fig. 4. Optimized decagonal monopole: (a) simulation model geometry with waveguide port, and (b) simulated reflection coefficient $|S_{11}|$ of the optimized antenna with resonance markers over the 2.16–15.73 GHz operating band.

7. Metamaterial (Metasurface) Integration and S_{11} Analysis

A 6×5 double-SRR metasurface on copper-backed 1.6 mm FR-4 is positioned at an air gap of 3.9 mm behind the antenna (Fig. 5(a)). While a conventional metal plate at this sub-4 mm spacing causes destructive phase cancellation below 9.6 GHz, the AMC metasurface provides in-phase reflection ($\phi_R - 2k_0h \approx 0$). Full-wave CST simulations verify that near-field metasurface loading preserves $|S_{11}| \leq -10$ dB across the operating band while substantially enhancing forward broadside gain (Fig. 5(b)).

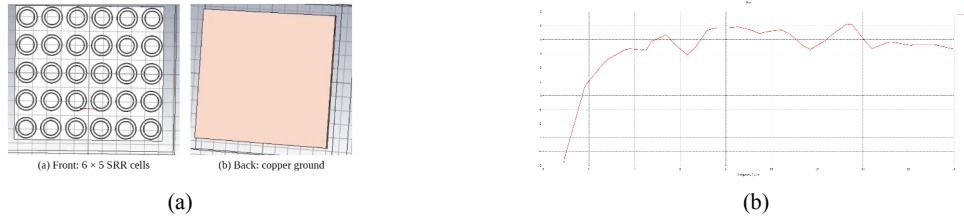


Fig. 5. Metasurface integration: (a) CST model of the 6×5 double-SRR metasurface array reflector, and (b) simulated standalone antenna IEEE gain across frequency (2.9–4.9 dBi over 3.1–10.6 GHz, providing the baseline for metasurface gain enhancement).

Work Done Till Mid-Semester

During Phase I, the single-element CPW decagonal monopole antenna was designed, optimized through parametric sweeps, and finalized in CST Studio Suite 2019. The 6×5 double-SRR metasurface array was modeled and its loaded S_{11} performance evaluated.

Table 2. Simulated performance summary of the optimized antenna (standalone).

Parameter	Simulated Value	Project Specification	Compliance
Impedance Bandwidth ($ S_{11} \leq -10$ dB)	2.16–15.73 GHz	2.1–15 GHz wideband	Exceeds
Fractional Bandwidth (FBW)	151.7 %	≥ 109 %	Exceeds
Resonances	2.73, 4.78, 9.23, 14.32 GHz	Wideband multi-mode	Achieved (4 resonances)
Deepest Return Loss	32.65 dB (at 2.73 GHz)	≥ 10 dB	Exceeds
IEEE Gain, 3.1–10.6 GHz	2.9 to 4.9 dBi (peak 5.09 dBi)	Baseline for AMC enhancement	Achieved
Metasurface Profile	$h = 3.9$ mm ($0.04\lambda_0$ at 3.1 GHz)	Low profile (< 5 mm)	Achieved

Challenges Encountered and Addressed

- **Broadband Sweep Numerical Artefacts:** Coarse frequency sweeps showed non-physical steps near 4.0 and 6.2 GHz; refined mesh passes resolved these.
- **Near-Field Metasurface Loading:** At an ultra-thin spacing of $h = 3.9$ mm, reactive loading slightly alters impedance; re-tuning L_g preserves matching.
- **Simulation Runtimes:** Composite 3D structures require large meshes; unit cells were characterized with Floquet ports prior to full-array verification.

Work Plan for Next Phase

In the next phase of the project, the primary focus will be extending the optimized wideband single-element antenna into a multi-port MIMO configuration and conducting experimental validation. The individual decagonal radiators will be arranged in an orthogonal orientation on a shared coplanar ground plane to exploit pattern and polarization diversity, thereby suppressing mutual coupling between adjacent antenna elements. The split-ring-resonator metasurface array will be expanded to serve as a common low-profile reflective backing for the multi-element system, enhancing forward directivity while preserving inter-element decoupling. Comprehensive diversity performance analyses will be carried out by evaluating critical MIMO parameters, including inter-port isolation, envelope correlation coefficient, diversity gain, total active reflection coefficient, channel capacity loss, and mean effective gain. Following electromagnetic simulation and layout refinement, physical prototypes will be fabricated on FR-4 substrates using standard photolithographic etching. Experimental characterization will be conducted using a calibrated vector network analyzer for multi-port reflection and transmission measurements, alongside anechoic chamber testing for far-field radiation patterns and realized gain.

Expected Outcomes

- A validated CPW-fed decagonal monopole antenna covering 2.16–15.73 GHz (151.7 % FBW), fully satisfying wideband requirements from 2.1 GHz to 15 GHz with IEEE gain of 2.9–4.9 dBi.
- A low-profile 6×5 double-SRR metasurface reflector operating at an air gap of $h = 3.9$ mm, substantially increasing forward broadside gain without impedance detuning.
- A rigorous comparative benchmark confirming that the AMC metasurface significantly outperforms a metal plate at identical spacing.
- A high-performance multi-port orthogonal MIMO antenna array providing high isolation and low envelope correlation across the wide operating band.
- Experimental validation via fabricated prototypes, calibrated VNA S-parameter measurements, and anechoic chamber far-field radiation testing.

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